CTL DECISION SHEET (DSH)

Standard(s) (incl. year)	Subclause(s)	Tracking No.	Year
OD-5014	Click here to enter text.	DSH 2111	2018
Category			
General			
Subject	**Keywords**	**Developed by**	**Approved at**
Clarification of "accuracy" in OD-5014:2016	Accuracy	WG1	2020 CTL Plenary Meeting

Question

OD-5014 specifies "default instrument **accuracies** for measurement ranges" (cl. 2.1).

It is unclear how to treat the term "accuracy" as there may be more than one possibility.

The term "accuracy" is specified in IEV 311-06-08:
accuracy (of a measuring instrument)
quality which characterizes the ability of a measuring instrument to provide an **indicated value** close to a true value of the measurand
Note 1 – This term is used in the "true value" approach.
Note 2 – Accuracy is all the better when the indicated value is closer to the corresponding true value.

However, it might be – as the document historically was created about 30 years ago – actually the "accuracy class" (IEV 311-06-09) is meant:
category of measuring instruments, all of which are intended to comply with a set of specifications regarding uncertainty

The latter seems to be used as some laboratories referencing to "full scale tolerance" (in the context with OD-5014) of a measuring instrument instead of the reading as indicated in IEV 311-06-08.

Which approach is correct?

It is suggested to apply the term "accuracy" – irrespective of meter technology - as indicated in IEV 311-06-08 (true value approach), so that each single value is measured with a tolerance not exceeding the "instrument accuracy of range" limits as indicated in OD-5014.

Decision
The values given in OD 5014 are intended to be accuracy of the full ranges indicated in the document. The accuracies provided are the agreed-upon default accuracies that are to be applied when the test standard does not state criteria for equipment accuracy. When the test standard requires a specified accuracy to improve repeatability of the test result, or where the default values in OD 5014 may not be considered sufficient by the standards developers, criteria for particular equipment would need to be included in the test standard.
Explanatory notes
(DSH 251 was replaced by OD 5014) Altering OD 5014 must have basis from both a technical and economic perspective. Where there are technical concerns, the first approach is to consider incorporating accuracy requirements in the particular test standard. A wide-based approach of altering OD 5014 without specific purpose may not provide benefit and may create economic impact.

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